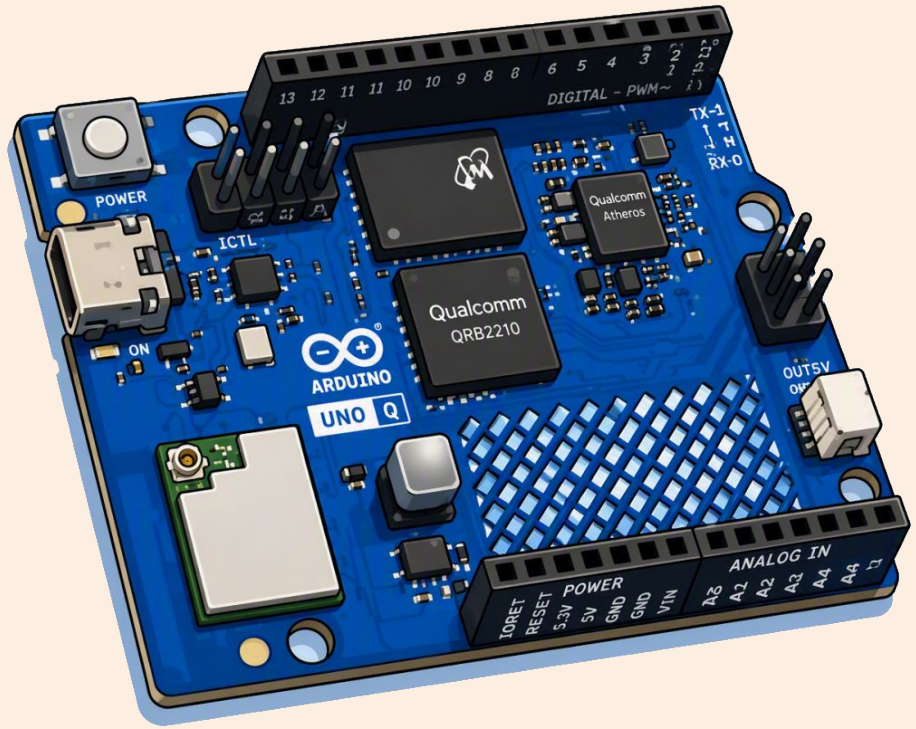
CHAPTER 1 · IESTI05 EDGE AI ENGINEERING

Setting Up the Arduino UNO Q

A terminal-first workflow: ADB, Wi-Fi, SSH

> Debian Linux + Zephyr RTOS
> no GUI · no monitor · no keyboard

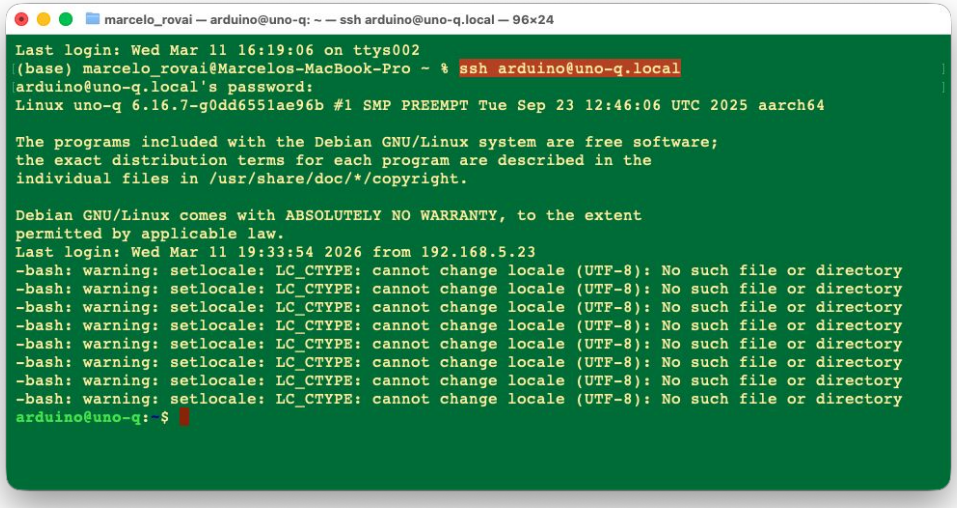
Marcelo Rovai · UNIFEI



WHERE WE ARE GOING

By the end of this chapter you will be here.

An SSH session on the board and a dual-brain app running from one command.



THE ROUTE

Seven stages, one shell at the end

1	**The board**	Two processors, one UNO footprint, an RPC bridge between them.
2	**ADB**	One tool on your computer, talking over the USB-C cable.
3	**Flash**	Optional, but it makes a whole classroom behave the same way.
4	**Connect**	First Linux shell on the board — and change the password.
5	**Wi-Fi + SSH**	Cut the cable. From here the board is a network machine.
6	**Files**	scp for scripts, SFTP for week one, git for anything you iterate on.
7	**Run apps**	arduino-app-cli: one command drives both processors.

STAGE 1 OF 8

What the board actually is

1

Two processors, one UNO footprint, and an RPC bridge between them.

ARCHITECTURE

One board, two processors

MPU · MICROPROCESSOR UNIT

Qualcomm QRB2210

Quad-core Cortex-A53 @ 2.0 GHz

Adreno 702 GPU · dual ISP

Runs Debian Linux — Python, AI inference, networking.

MCU · MICROCONTROLLER UNIT

ST STM32U585

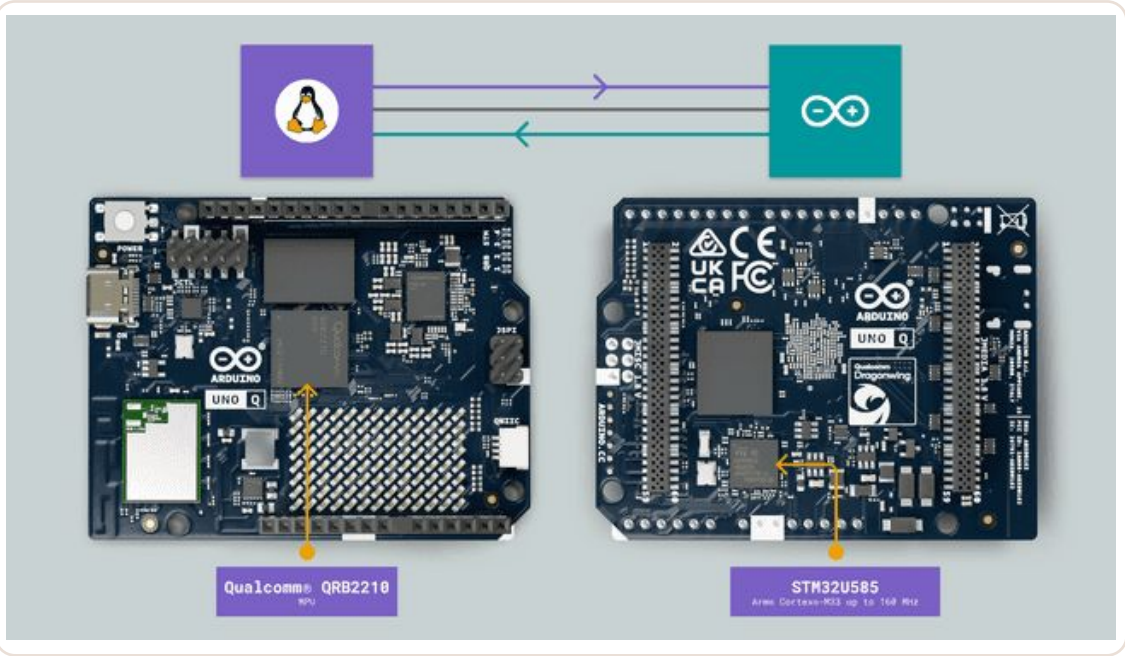
Cortex-M33 @ 160 MHz

Runs the Arduino Core on Zephyr — deterministic, real-time hardware control.

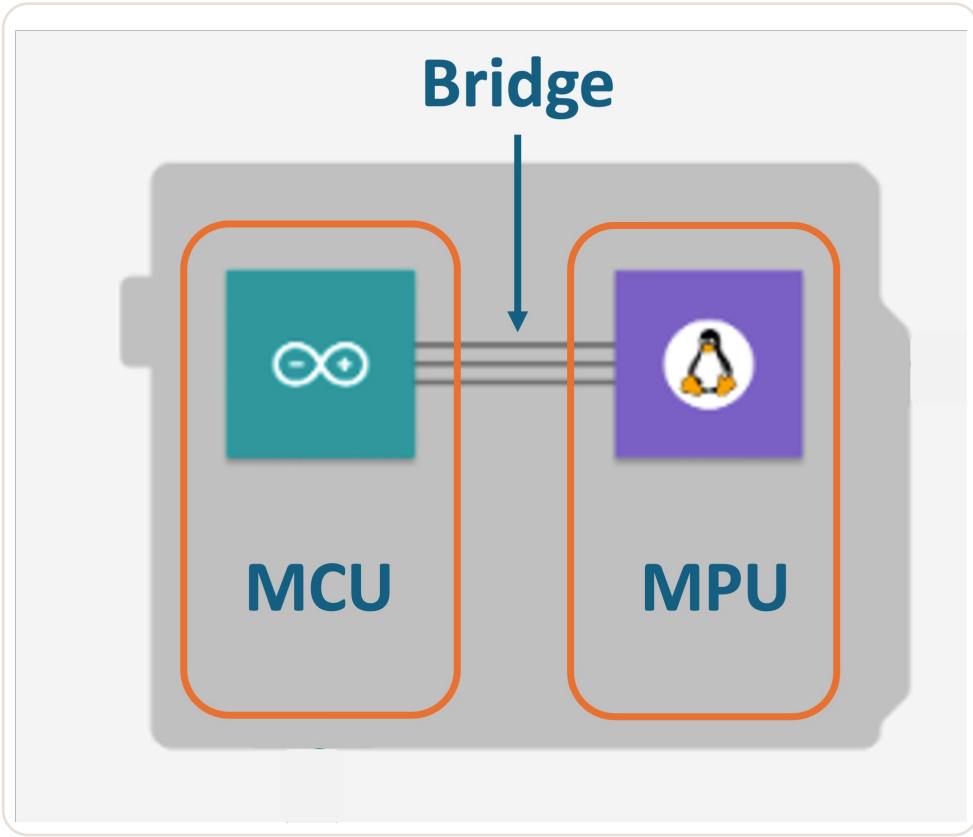
Bridge — Arduino's RPC library — lets Python on the MPU call functions in the sketch on the MCU, and back.

ARCHITECTURE

The bridge between them



UNO Q Architecture



Python on the MPU ⇄ Arduino sketch on the MCU

WHICH ONE TO BUY

Two variants, one architecture

ENTRY

2 GB

LPDDR4X · 16 GB eMMC

Headless SSH development, lightweight edge AI, TinyML.

COMFORTABLE

4 GB

LPDDR4X · 32 GB eMMC

SBC mode with a display, larger AI models, multitasking.

BOTH

Identical

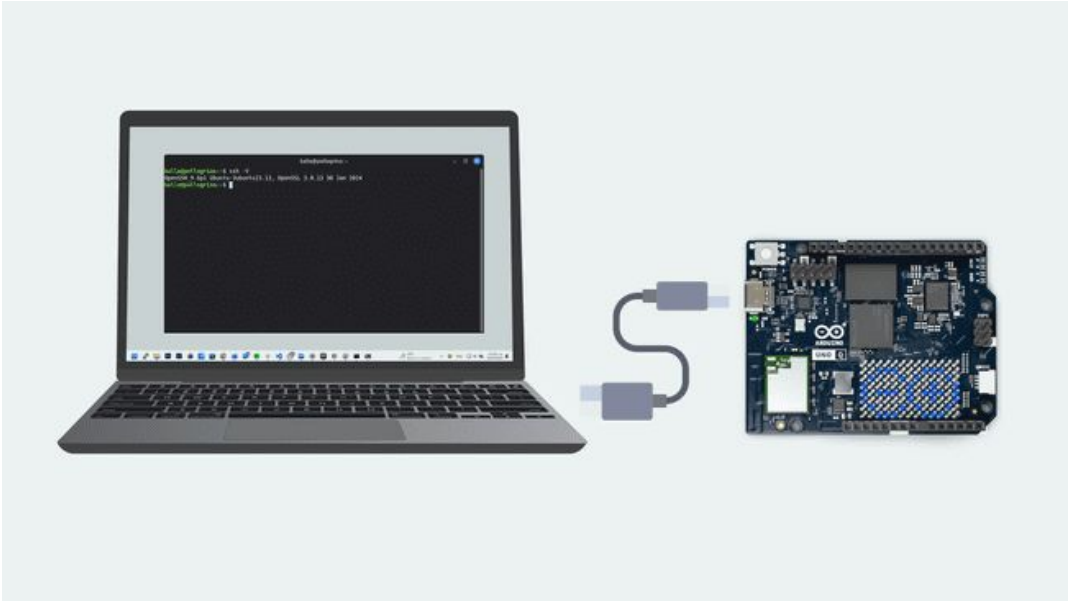
Same processors · Wi-Fi 5 + BT 5.1 · USB-C · UNO headers · Qwiic · 8×13 LED matrix.

RAM is the only difference that matters — and it determines whether generative AI is comfortable.

THE CABLE

The first thing that goes wrong is the cable.

One USB-C port carries power, data, and video. It must be a data cable — not a charge-only one. Some hubs and USB-C adapters are not recognized at all.



Accessing the board shell

APPROACH

Why terminal-first, and not the App Lab GUI

1

Full control

You work inside the board's Debian environment — nothing hidden behind an abstraction.

2

Code lives with you

Projects stay on your machine under Git, not locked inside the board.

3

Transferable skills

Linux, SSH and SCP are what real edge-AI deployments actually use.

4

Honest about failure

You see the actual error, not a spinner. That is where the learning is.

Also: scriptable, and editor-agnostic — VS Code, Vim, anything.

STAGE 2 OF 8

Host setup: ADB

2

One tool on your computer opens a shell on the board over the USB-C cable.

BEFORE YOU START

What you need

HARDWARE

On the bench

Arduino UNO Q (2 GB or 4 GB)

USB-C data cable

Host computer — Linux, macOS or Windows

NETWORK

Same one, both ends

A Wi-Fi network the board and your computer can both reach.

You will need the SSID and password.

HOST SOFTWARE

Three commands

adb — the first headless connection

ssh — remote access over Wi-Fi

scp — file transfer

INSTALL

Getting ADB onto your computer

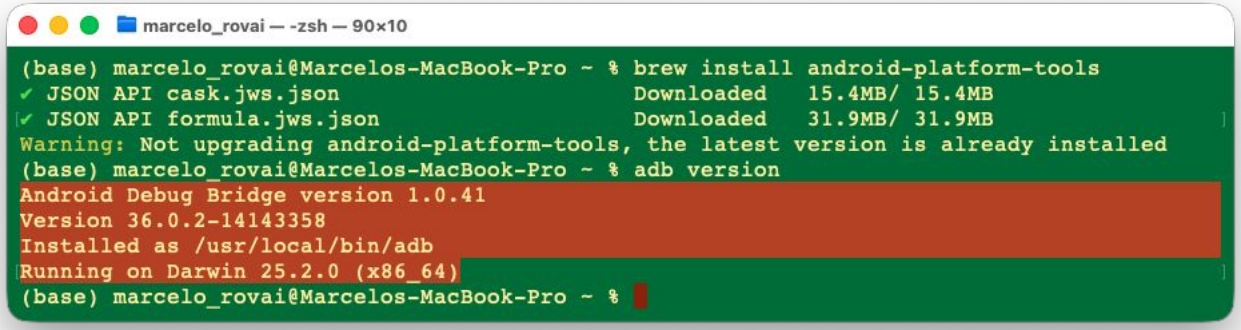
```
# Linux — Ubuntu / Debian
sudo apt update
sudo apt install android-tools-adb

# macOS — Homebrew
brew install android-platform-tools

# Windows
Download SDK Platform-Tools, extract to C:\platform-tools,
then add that folder to your PATH
```

VERIFY

Check ADB before you plug anything in

```
$ adb version
Android Debug Bridge version 1.0.41
```

> Run this with the board DISCONNECTED. If it fails here, nothing downstream will work — and you will spend an hour blaming the board.



STAGE 3 OF 8

Flashing the latest image

3

Optional — but essential if you want a whole classroom behaving the same way.

RECOMMENDED FIRST STEP

Why flash a board that already boots

1

Random restarts

The old image has no ZRAM compression — apps restart unpredictably under memory pressure.

2

ADB defaults to root

A security problem, and the wrong habit to learn in the first minute.

3

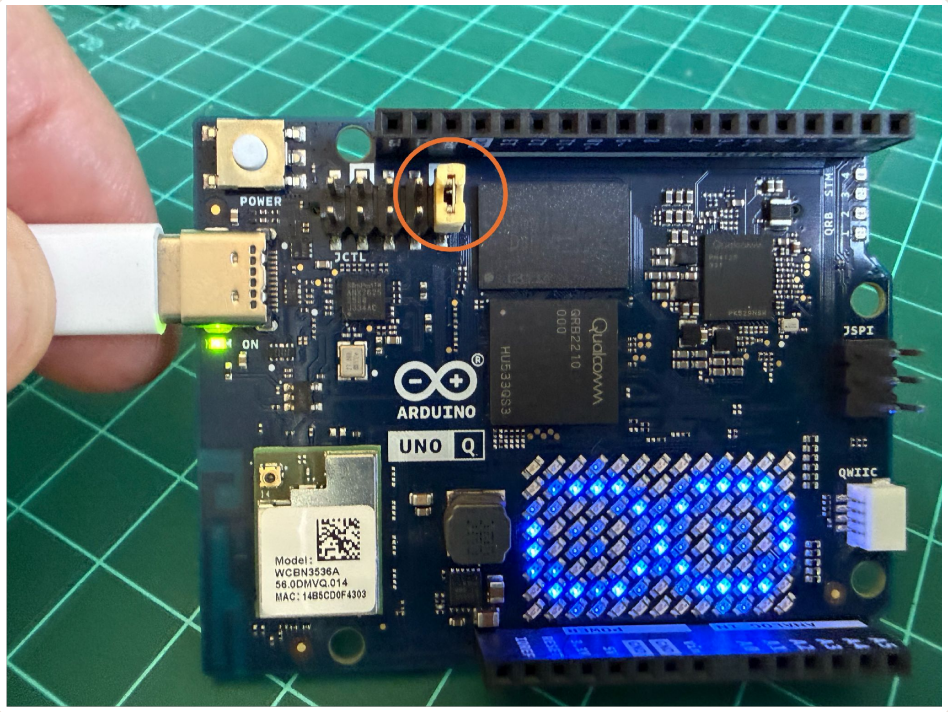
No HDMI audio

Simply missing from the shipped image.

> Flashing erases everything — the board goes back to factory state. The tool downloads ~2.3 GB, so use a good connection, or download once and share the image.

STEP 1 OF 3

Jumper the JCTL header — this is EDL mode



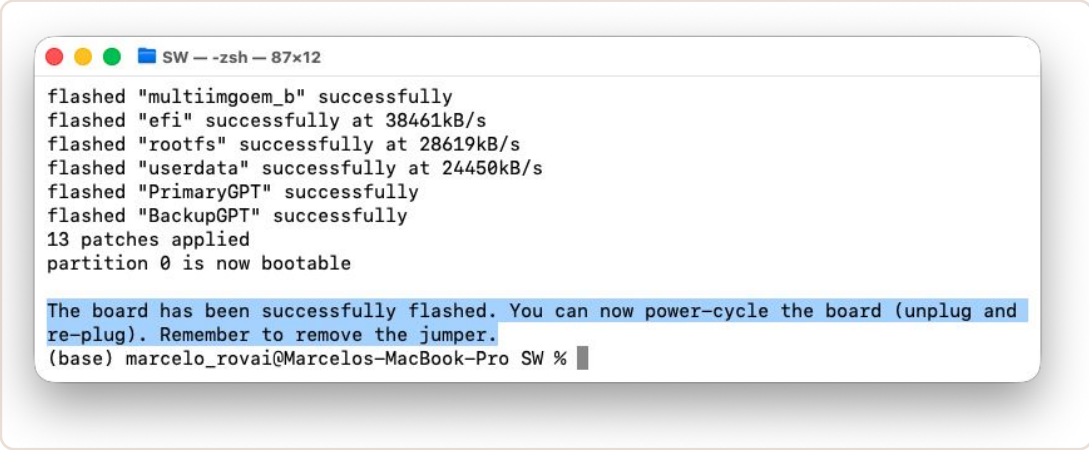
The LED matrix loops the Arduino logo — you are in EDL mode

Board powered OFF. Short the two pins FURTHEST from the USB-C connector, then plug the cable in. Jumper first, cable second.

STEP 2 OF 3

Run the flasher

```
$ ./arduino-flasher-cli flash latest
Download image? [y/N] y
Downloading ~2.3 GB ...
Flashing ...
```

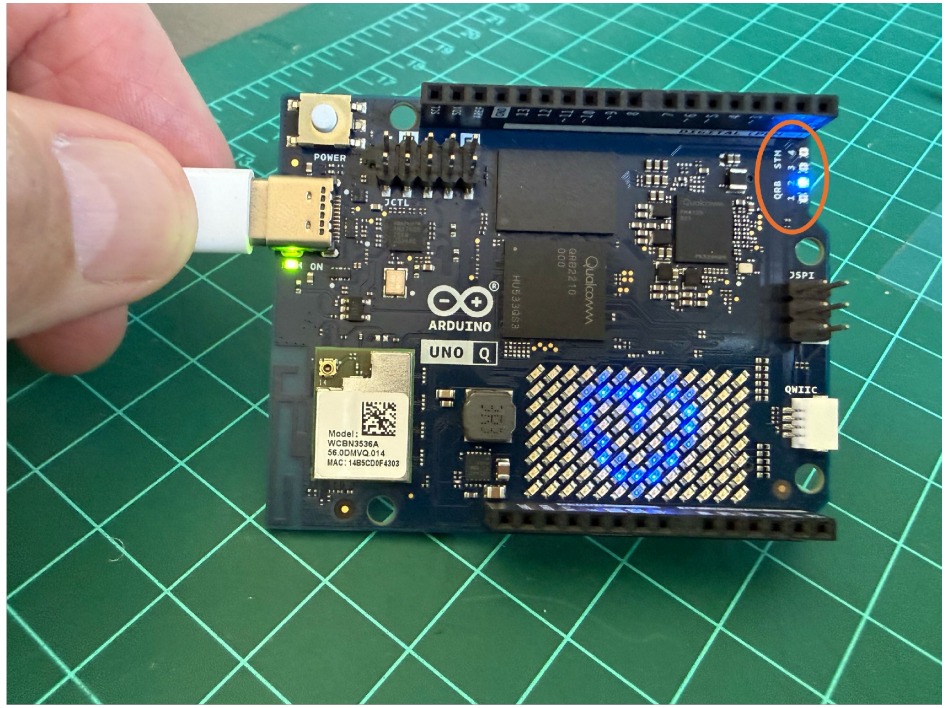


Flashing in progress

> Do not disconnect the cable or interrupt the process. Get arduino-flasher-cli from the Arduino Software Download page or its GitHub releases.

STEP 3 OF 3

Remove the jumper and reboot



Logo animation ends in a heart, blue LED 2 lit — the flash worked

Disconnect · remove the jumper · reconnect.

STAGE 4 OF 8

First connection over ADB

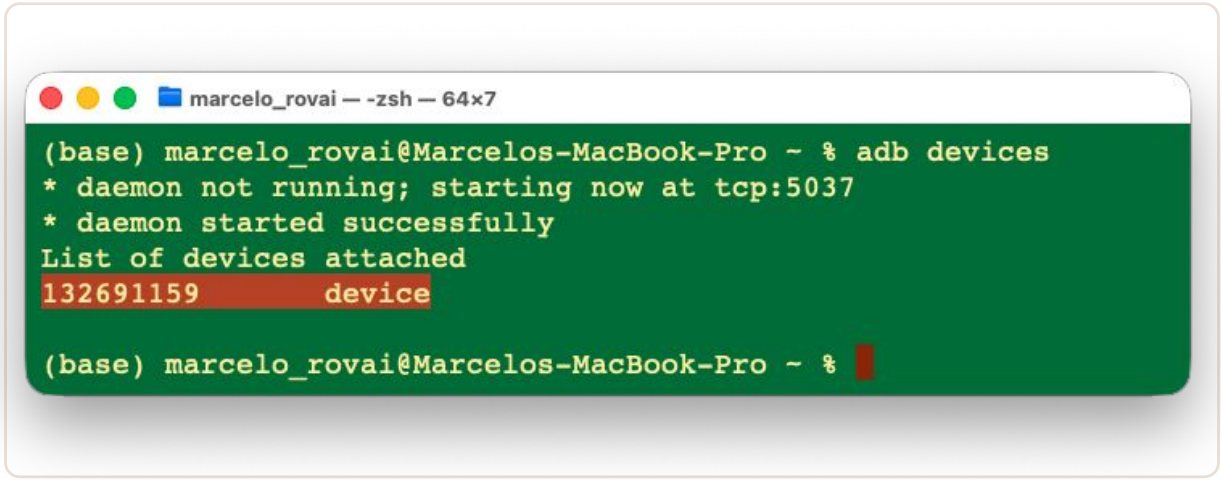
4

Plug in the cable, open a shell, change the password.

HEADLESS BRING-UP

Is the board there?

```
$ adb devices
List of devices attached
XXXXXXXX    device
```



adb devices — the board is seen

> Do not debug anything else until this line shows a device.

IF THE LIST COMES BACK EMPTY

Check these three, in this order

1

The cable

Is it a data cable, or a charge-only one? This is the answer about half the time.

2

The port

Plug directly into the computer. Hubs and USB-C adapters are frequently not recognised.

3

udev rules

On Linux only — without them the device is there but you have no permission to see it.

> Order matters: cable, then port, then udev. Working down the list beats guessing.

YOU ARE ON THE BOARD

Open a shell and change the password

```
$ adb shell
arduino@uno-q:~$

$ sudo passwd arduino
```

> Default credentials are arduino / arduino. Change the password NOW — you need the new one for SSH, and the board will keep nagging you until you do.

KNOW WHAT YOU ARE WORKING WITH

Four commands worth running once

```
$ cat /etc/os-release   # which Debian
$ lscpu                 # the four A53 cores
$ free -h               # RAM and swap
$ df -h                 # eMMC space left
```

> Run these on day one. Knowing the real numbers explains a lot of later behavior.

STAGE 5 OF 8

Wi-Fi and SSH

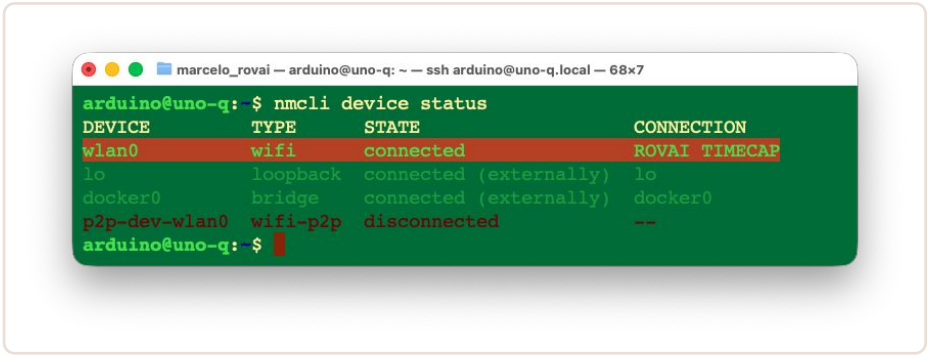
5

Configure the network from inside the ADB shell, then drop the cable.

NETWORKING

Join a Wi-Fi network with nmcli

```
# inside the ADB shell
$ nmcli dev wifi list

$ sudo nmcli dev wifi connect "YOUR_SSID" \
      password "YOUR_PASSWORD"
```



nmcli device status

> Quote the SSID if it contains spaces. Some networks hide their 5 GHz SSID — check the band.

NETWORKING

Confirm, and get the address you need

```
$ nmcli device status
wlan0    wifi    connected

$ hostname -I
192.168.5.85
```

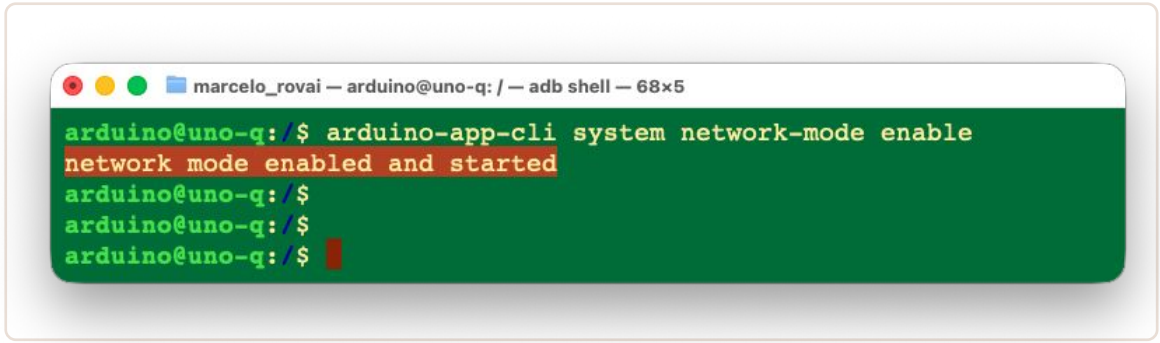
Connected — write this address down

> Better than writing it down: take the MAC from ip addr show wlan0 and give the board a static lease on your router, so the address survives reboots.

REMOTE ACCESS

Enable SSH — the step everyone forgets

```
# on the board, in the ADB shell
$ arduino-app-cli system network-mode enable
$ exit
```



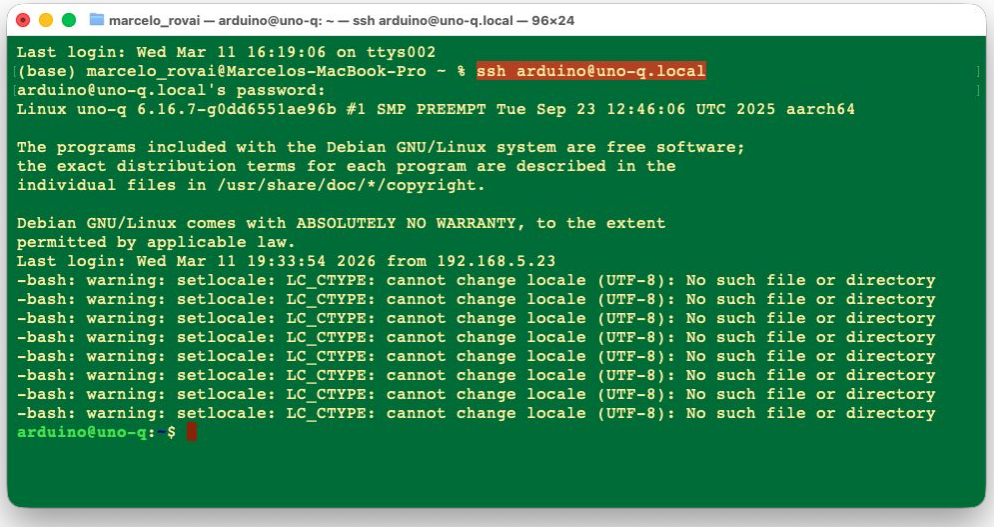
network-mode enable

> This is the single most common place people get stuck. SSH exists on the board, but it is not active until this command runs.

REMOTE ACCESS

Drop the cable

```
# from your host computer now
$ ssh arduino@192.168.5.85

# or, on networks with mDNS
$ ssh arduino@uno-q.local
```



Connected over Wi-Fi — the cable is just power now

> Password rejected? Go back in over ADB — adb shell , then sudo passwd arduino — exit, and retry.

HOUSEKEEPING

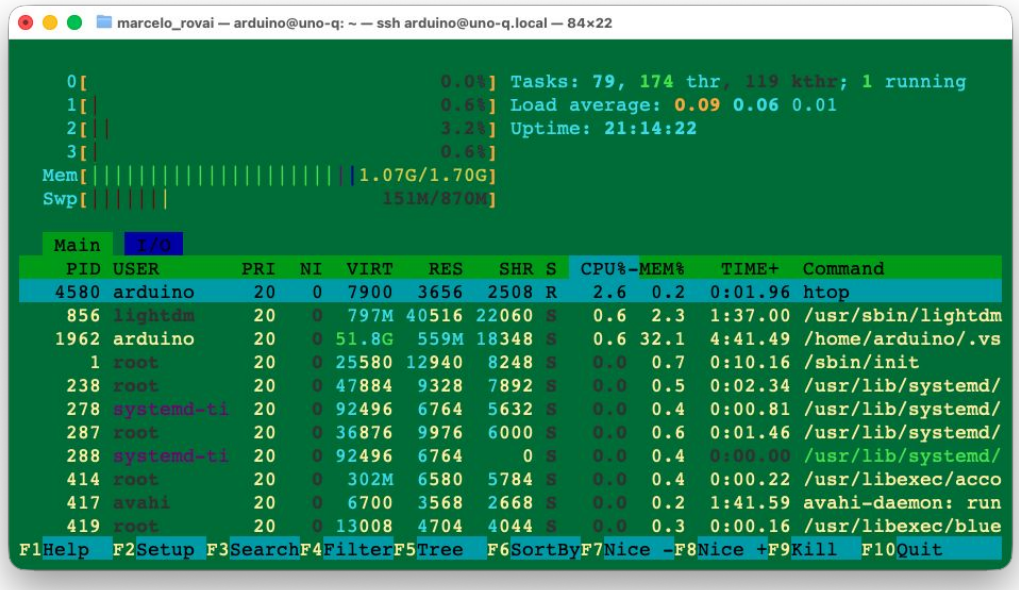
Update the system once

```
$ sudo apt update && sudo apt upgrade
$ sudo reboot
```

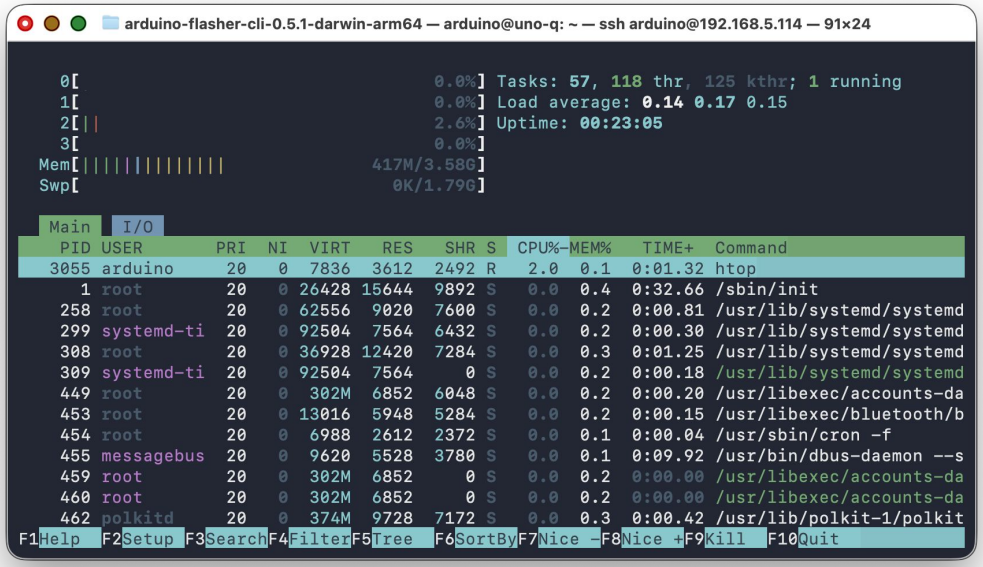
> Do this now, over Wi-Fi, before installing anything else.

KNOW YOUR BOARD

htop: 2 GB versus 4 GB, side by side



2 GB variant



4 GB variant

> Both the RAM and the swap are doubled on the 4 GB board.

STAGE 6 OF 8

Getting code onto the board

6

scp for scripts, SFTP for week one, git for anything you iterate on.

MOVING CODE

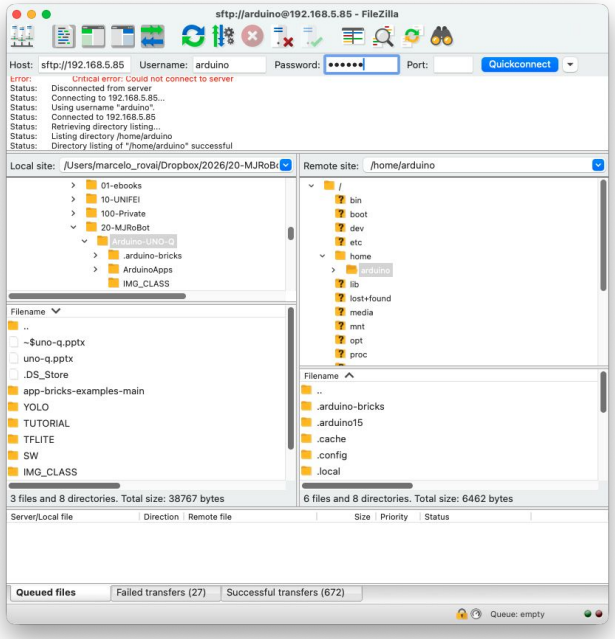
scp — the one that scripts

```
$ cd <your-project-folder>
$ scp -r * arduino@192.168.5.85:~/ArduinoApps/<folder>/
```

> Because it is a plain command, it drops straight into a Makefile or a deploy script. For repeated syncs use rsync — it only sends what changed, which matters over Wi-Fi.

MOVING CODE

SFTP — the one they will actually use



FileZilla: both filesystems side by side

Host field: sftp://192.168.5.85 · board credentials · Quickconnect.

STAGE 7 OF 8

Running things: arduino-app-cli

7

One command orchestrates the Python container and the MCU sketch.

AN IMPORTANT DISTINCTION

Two classes of app

READ-ONLY, CLI-MANAGED

examples:*

Live in /var/lib/arduino-app-cli — not in your home directory, which is why ls never shows them.

arduino-app-cli app start examples:blink

YOURS, EDITABLE

user:*

Live in ~/ArduinoApps/. The only place ls shows anything, and the only place you edit.

arduino-app-cli app new "Blink"

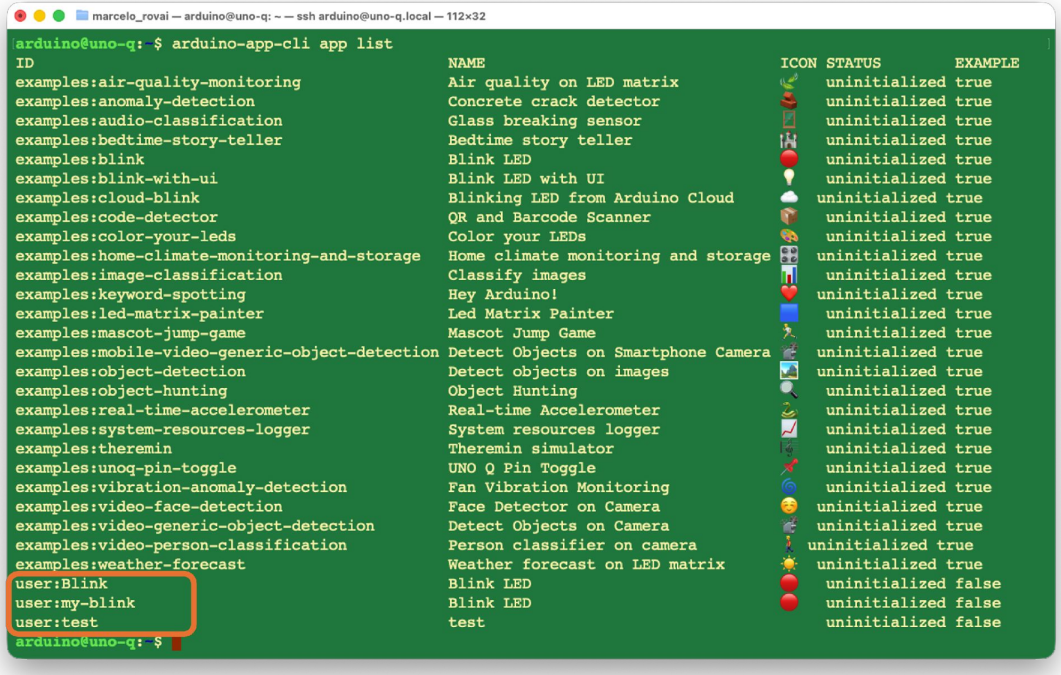
This trips everyone up: app list shows examples you cannot find with ls. They are managed by the CLI.

BEFORE YOU EXPLORE

Pull the latest components

```
$ arduino-app-cli system update
```

> This takes a long time. Start it before a break, not in the middle of a demo.



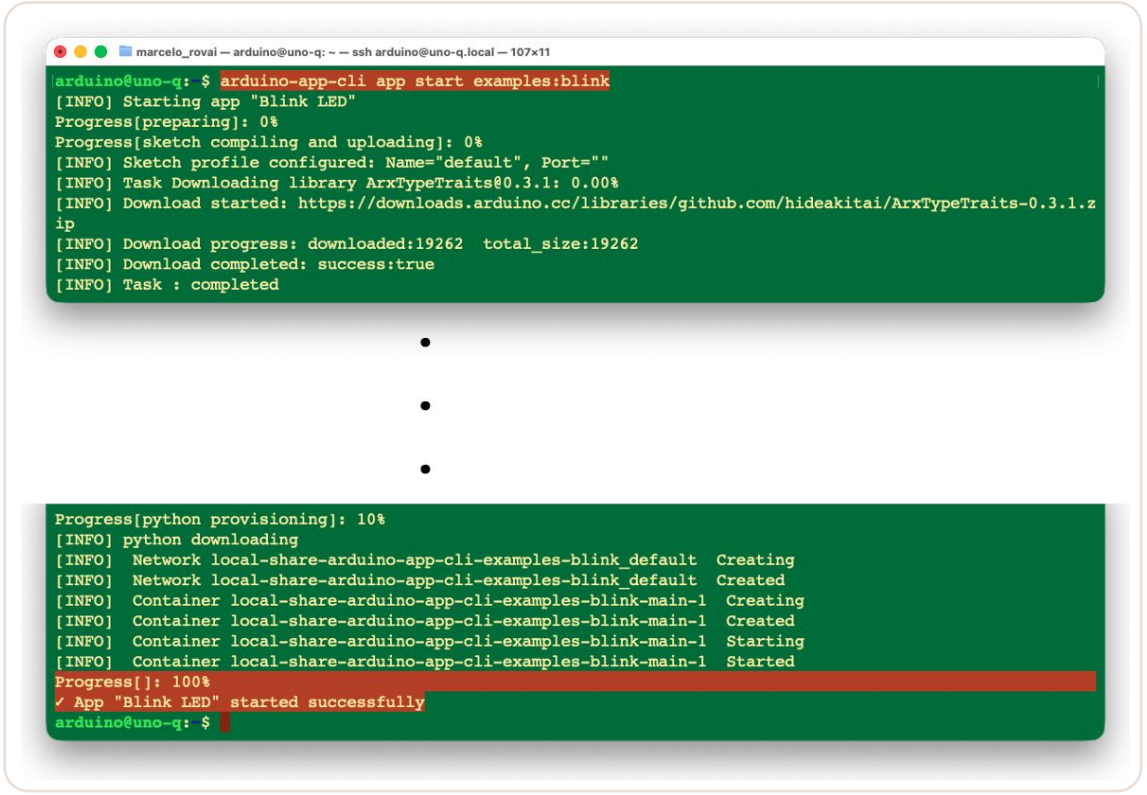
app list — both classes, one listing

THE FIRST DUAL-BRAIN APP

Start it

```
$ arduino-app-cli app start examples:blink
```

> The FIRST run takes several minutes — it downloads Arduino libraries and builds the Python container. Warn people, or half of them will decide it is broken and reboot.



Starting the app

THE FIRST DUAL-BRAIN APP

Watch it, then stop it

```
$ arduino-app-cli app logs
examples:blink
$ arduino-app-cli app stop
examples:blink
```

app logs — the Python side talking

> Add --follow to tail the log live.

REFERENCE

The arduino-app-cli commands you will actually use

Command	What it does
`arduino-app-cli app new "<name>"`	Create a new app skeleton in ~/ArduinoApps/
`arduino-app-cli app start <path>`	Build if needed, then start the app
`arduino-app-cli app logs <path>`	View the Python-side log — add --follow to tail it
`arduino-app-cli app stop <path>`	Stop a running application
`arduino-app-cli app list`	List installed and running apps, both classes
`arduino-app-cli system update`	Update the CLI and board components (slow)
`arduino-app-cli system cleanup`	Remove unused containers — reclaim eMMC space

> app start compiles the sketch, flashes the MCU, builds the Python environment and launches the app — both processors, one command.

SURVIVAL KIT

The Linux commands this course leans on

SYSTEM

```
cat /etc/os-release
lscpu
free -h
df -h
htop
```

NETWORK

```
ip addr
nmcli device status
nmcli dev wifi list
hostname -I
```

PACKAGES

```
sudo apt update
sudo apt upgrade
sudo apt install <pkg>
sudo apt autoremove
```

FILES

```
ls -lah
cd / mkdir / cp / mv
rm -rf <dir>
cat / nano <file>
```

SERVICES

```
sudo systemctl status ssh
sudo systemctl restart ssh
sudo reboot
sudo shutdown -h now
```

CLEANUP

```
df -h
sudo apt clean
arduino-app-cli system
cleanup
docker system prune -a
```

Standard Debian — anything you know from a Raspberry Pi or a cloud VM applies here unchanged.

STAGE 8 OF 8

When it breaks — and what it is for

8

The failures you will meet in week one, then an honest look at the board's limits.

WEEK ONE

The failures you will actually hit

1

ADB does not see the board

Charge-only cable, a hub in the path, or missing udev rules on Linux.

2

SSH connection refused

network-mode enable was never run — or you are on a different network.

3

SSH password rejected

Reset it over ADB: adb shell , then sudo passwd arduino .

4

Everything is mysteriously slow

Check htop. If swap is in use, close something — the A53s do not recover from thrashing.

> Nearly all of these are cable, network, or memory. Teach the diagnosis order and they stop being interesting.

BE REALISTIC

What this board is not good at

COMPUTE

Roughly a Pi 3

Cortex-A53 cores. Models beyond ~2B parameters are impractical. For heavy inference a Pi 5 or an accelerator is still the right answer.

MEMORY & PORTS

Tight, and singular

2 GB is tight for editor + container + model. One USB-C port shares power, data and video — SBC mode needs a hub.

SILICON & SOFTWARE

No NPU, pre-1.0 tools

Inference runs on CPU and GPU. App Lab and the CLI are pre-1.0 — expect rough edges. No built-in camera.

State the ceiling on day one: students who know it design within it, instead of blaming themselves.

CARRY THESE OUT OF CHAPTER 1

Three things

1

The board is two computers

An MPU running Debian and an MCU running Zephyr, talking over Bridge. Knowing which side a problem is on is half of debugging it.

2

The terminal is the interface

ADB to bootstrap, Wi-Fi to cut the cord, SSH from then on. Code lives on your machine under Git; the board just runs it.

3

One CLI drives both processors

app new, app start, app logs, app stop. Every chapter from here uses exactly those four.

RESOURCES

Where to read further

Resource	Where
UNO Q documentation	docs.arduino.cc/hardware/uno-q
UNO Q datasheet (PDF)	docs.arduino.cc/resources/datasheets/ABX00162-datasheet.pdf
Arduino App CLI tutorial	docs.arduino.cc/software/app-lab/tutorials/cli
Arduino App CLI (GitHub)	github.com/arduino/arduino-app-cli
Edge Impulse — UNO Q	docs.edgeimpulse.com/hardware/boards/arduino-uno-q
arduino-flasher-cli releases	github.com/arduino/arduino-flasher-cli/releases

Next — Ch. 2, Generative AI at the Edge: building llama.cpp from source, running Qwen3.5 from the CLI and a local llama-server, and calling it from Python.

Questions?

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